

Course	ENCN423/ENCN627 - Sustainable Energy Systems	 UC UNIVERSITY OF CANTERBURY <i>Te Whare Wānanga o Waitaha</i> CHRISTCHURCH NEW ZEALAND
Assessment	Test 3 - Practice	
Topic	Solar, Batteries, Transport, Fuels & Industry	
Duration	1.5 hours	

Instruction:

- Answer the following questions as completely as possible and show all your work (where appropriate).
- You may ask for clarification on question text.
- A formula sheet can be found at the back of this test booklet
- Submit your answer booklet and test sheet at the end of the test.

TEST BEGINS ON PAGE 2

Multiple choice questions [2 pts each]

1. What are the advantages behind half-cut PV cells as opposed to full cells? [multiple correct answer]
 - (a) **Lower internal electrical resistance losses.**
 - (b) **Improved performance under partial shading conditions.**
 - (c) **Reduced risk of hot-spot formation and related cell degradation.**
 - (d) Lower manufacturing costs per watt due to simpler processing steps.

2. Which of the following statements accurately describe the physical processes and components that allow a solar PV system to function? [multiple correct answer]
 - (a) **The p-n junction creates an internal electric field that separates photo-generated electrons and holes, directing them to opposite sides.**
 - (b) **The photovoltaic effect occurs when photons with enough energy strike the semiconductor, exciting electrons so they can flow as current.**
 - (c) In the p-n junction, solar energy (photons) hit the atomic nuclei to release free electrons.
 - (d) **An inverter uses pulse-width modulation (rapid switching) along with inductors and capacitors to transform DC into a smooth AC sine wave.**

3. What are the four major components that make up a standard Lithium-ion battery cell? [multiple correct answers]
 - (a) Inverter
 - (b) **Anode**
 - (c) Capacitor
 - (d) **Cathode**
 - (e) **Electrolyte**
 - (f) **Separator**

4. Why is lithium used as the primary active material in most modern high-performance batteries compared to other metals?
 - (a) **It has a very high electrochemical potential and the lowest atomic mass of all metals, enabling high energy density.**
 - (b) It is the most chemically stable metal, preventing internal degradation.

- (c) It has the highest electrical conductivity of all metals, minimizing internal resistance.
 - (d) It is the most abundant and cheap metal in the Earth's crust, making it the most cost-effective option.
5. A passenger electric vehicle (EV) is travelling at a constant speed on a highway. Which modification would most effectively reduce energy consumption at high speeds?
- (a) Reducing vehicle mass by 10%
 - (b) Reducing rolling resistance coefficient by 10%
 - (c) **Reducing drag coefficient by 10%**
 - (d) Increasing battery capacity by 10%
6. Two marine fuels have the same gravimetric fuel consumption, but different lower heating values and densities. Which parameter most directly determines the total fuel energy required per ton-kilometer?
- (a) Fuel density and lower heating value
 - (b) **Lower heating value only**
- Think of it this way: your engine doesn't care how much space the fuel takes up in the tank (density); it only cares about the total chemical energy available in the mass of fuel it burns to move the ship. Since our consumption here is already measured by mass, density is just extra information that doesn't change the amount of energy the engine must extract from that mass to get the job done.
- (c) Fuel density only
 - (d) Tank-to-wheel (TTW) efficiency only
7. Which fuels below require carbon during synthesis?
- (a) Hydrogen only
 - (b) Ammonia only
 - (c) **Methanol and Fischer–Tropsch liquids**
 - (d) Hydrogen and ammonia only
8. What is the main purpose of Fischer–Tropsch synthesis?
- (a) Produce electricity directly from water
 - (b) **Produce liquid hydrocarbons from hydrogen and carbon**

- (c) Split ammonia into nitrogen and hydrogen
 - (d) Compress hydrogen into liquid form
9. What is the primary technical barrier when replacing conventional Coal/Coke-based Blast Furnace-Basic Oxygen Furnace (BF-BOF) systems with Hydrogen-based Direct Reduced Iron Electric Arc Furnace (H2-DRI-EAF) systems?
- (a) Hydrogen molecules are too small and can go right through the iron ore, resulting in slow reaction kinetics compared to carbon monoxide.
 - (b) **The reduction of iron ore with hydrogen requires massive amounts of renewable electricity and high-temperature heat.**
 - (c) Hydrogen-based reduction prevents the melting of iron, making it impossible to separate slag from the pure liquid metal.
 - (d) Electric Arc Furnaces (EAF) are chemically incompatible with iron pellets and can only process recycled scrap steel.
10. Why is switching the thermal energy source of cement production to 100% renewable electricity or green hydrogen insufficient on its own to achieve net-zero emissions?
- (a) **The majority of a cement plant's CO₂ emissions are “process emissions” generated directly by the chemical calcination of limestone.**
 - (b) Electric arc heating cannot reach the temperatures above 1400 °C required for clinker formation in the rotary kiln.
 - (c) Hydrogen combustion yields water vapor, which chemically deactivates the calcium silicate compounds needed for cement strength.
 - (d) Renewable energy systems cannot supply the high voltage required to operate the heavy raw mills and clinker grinders.

Calculation questions

11. Sizing a rooftop PV system

- (a) Calculate the average peak solar hours (PSH) per day given the following irradiance data of accumulated insolation (hours)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
200	170	145	115	100	80	80	85	95	120	140	190

$$Insolation_{total} = 200 + 170 + 145 + \dots + 140 + 190 = 1520 \text{ hours/year}$$

$$Insolation_{day} = \frac{1520 \text{ hours/year}}{365 \text{ days/year}} = 4.2 \text{ hours}$$

- (b) Using your answer from (a), calculate the number of 550 W PV panels required for a house to generate 6650 kWh/yr, assuming a performance ratio of 90%

$$E = \frac{6650 \text{ kWh/year}}{365 \text{ days/year}} = 18.2 \text{ kWh/day}$$

$$P_{array} = \frac{E}{PSH \cdot PR} = \frac{18.2}{4.2 \cdot 90\%} = 4.8 \text{ kW}$$

$$N_{panel} = \frac{P_{array}}{P_{module}} = \frac{4800 \text{ W}}{550 \text{ W}} = 8.7 = 9 \text{ panels}$$

- (c) Calculate the minimum and maximum string size for these panels given the following data:

- $V_{MPP} = 191.4 \text{ V}$
- $V_{OC} = 218.1 \text{ V}$
- $V_{Inv-min} = 295 \text{ V}$
- $V_{Inv-max} = 875 \text{ V}$
- $T_{ambient-max} = 37^\circ\text{C}$
- $T_{ambient-min} = -2^\circ\text{C}$
- $T_{add} = 30^\circ\text{C}$
- $T_k = -0.30\%/^\circ\text{C}$
- $T_{STC} = 25^\circ\text{C}$

$$V_{MPP-min} = 191.4 \cdot \left[1 + \left((37 + 30 - 25) \cdot \frac{-0.30}{100} \right) \right] = 158.5 \text{ V}$$

$$V_{OC-max} = 218.1 \cdot \left[1 + \left((-2 - 25) \cdot \frac{-0.30}{100} \right) \right] = 235.8 \text{ V}$$

$$N_{string-min} = \frac{V_{Inv-min}}{V_{MPP-min}} = \frac{295}{158.5} = 1.86 = 2 \text{ panels}$$

$$N_{string-max} = \frac{V_{Inv-max}}{V_{OC-max}} = \frac{875}{235.8} = 3.7 = 3 \text{ panels}$$

12. A commercial building in Christchurch intends to install a 120 kW solar PV system. The site has an average annual 1400 PSH/year.

- CAPEX: \$1400/kW
- Annual OPEX: \$ 25/kW
- Lifetime: 25 years
- Discount rate: 6%
- Performance ratio: 85%

(a) Calculate the annual electricity output (kWh/year) generated by the system.

$$120 \text{ kW} \cdot 1400 \text{ PSH/year} \cdot 85\% = 142\,800 \text{ kWh}$$

(b) Calculate the Levelized Cost of Electricity (LCOE) in \$/MWh.

$$crf = \frac{6\% \cdot (1 + 6\%)^{25}}{(1 + 6\%)^{25} - 1} = \frac{0.2575}{3.29} = 7.8\%$$

$$LCOE = \frac{1400 \cdot 7.8\% + 25}{1400 \cdot 85\%} = 113 \text{ $/MWh}$$

or

$$LCOE = \frac{1400 \cdot 120 \cdot 7.8\% + 25 \cdot 120}{1400 \cdot 120 \cdot 85\%} = 113 \text{ $/MWh}$$

13. An off-grid telecom station requires a daily useful energy dispatch of 48 kWh from a Li-ion battery bank to cover night-time loads.

- Nominal voltage: 48 V
- Maximum allowed depth of discharge (DoD): 80%
- Inverter efficiency: 92%
- Battery charging/discharging efficiency: 95%

(a) Calculate the minimum total nominal capacity required for the battery in Ampere-hours (Ah) to satisfy the daily load while protecting the DoD limits and accounting for inverter losses.

$$E_{supply} = \frac{E_{demand}}{\eta_{inverter} \cdot \eta_{discharge}} = \frac{48 \text{ kWh}}{92\% \cdot 95\%} = 54.9 \text{ kWh}$$

$$E_{stored} = \frac{E_{supply}}{DoD_{max}} = \frac{54.9 \text{ kWh}}{80\%} = 68.6 \text{ kWh}$$

$$C_{battery} = \frac{E_{stored}}{V_{nominal}} = \frac{68\,600\,Wh}{48\,V} = 1430\,Ah$$

- (b) If the battery bank is charged exclusively by solar PV during the day, calculate the total DC electricity (kWh) the PV array must supply into the battery to fully replenish the discharged load, accounting only for losses.

$$E_{PV} = \frac{E_{supply}}{\eta_{charge}} = \frac{54.9\,kWh}{95\%} = 57.8\,kWh$$

14. An industrial site has just installed a battery energy storage system (BESS) with a capacity of 950 Ah, which can operate between 400 V and 600 V. In order to preserve the longevity of the BESS, it will never be discharged below 15% of its capacity, or charged above 90%. Calculate the useful energy stored in this BESS

$$E_{useful} = C \cdot V_{nominal} \cdot (SoC_{max} - SoC_{min}) = 950\,Ah \cdot \frac{600\,V + 400\,V}{2} \cdot (90\% - 15\%) = 356\,250\,Wh$$

15. An electric SUV accelerates from 40 km/h to 100 km/h in 10 seconds on a flat road. The vehicle has:

- Rolling resistance coefficient: $C_{roll} = 0.011$
- Aerodynamic drag coefficient: $C_{drag} = 0.26$
- Cross-sectional area: $A = 2.6\,m^2$
- Air density: $\rho = 1.225\,kg/m^3$

The total energy required for the acceleration event is 780 kJ. Assume constant acceleration.

- (a) Calculate the total vehicle mass.

$$40\,km/h = 11.1\,m/s$$

$$100\,km/h = 27.8\,m/s$$

$$E_{total} = \Delta KE + W_{roll} + W_{drag} = 780\,kJ$$

$$\Delta KE = \frac{1}{2} \cdot m \cdot (27.8^2 - 11.1^2) = 324.8 \cdot m$$

$$W_{\text{roll}} = 0.011 \cdot m \cdot 9.81 \cdot \left(\frac{11.1 + 27.8}{2} \cdot 10 \right) = 21 \cdot m$$

we use

$$W_{\text{drag}} = \frac{1}{8} \cdot \rho \cdot C_{\text{drag}} \cdot A \cdot \frac{v_2^4 - v_1^4}{a}$$

which applies to the work done against aerodynamic drag when an object is undergoing **a constant acceleration** over a specific interval of speeds.

It is a variation of

$$W_{\text{drag}} = \frac{1}{2} \cdot \rho \cdot C_{\text{drag}} \cdot A \cdot v^2$$

which applies when an object is moving at **a constant velocity**

$$a = \frac{27.8 - 11.1}{10} = 1.67$$

$$W_{\text{drag}} = \frac{1}{8} \cdot 1.225 \cdot 0.26 \cdot 2.6 \cdot \frac{27.8^4 - 11.1^4}{1.67} = 36\,081 \, J \approx 36.1 \, kJ$$

finally

$$E_{\text{total}} = \Delta KE + W_{\text{roll}} + W_{\text{drag}} = 324.8 \cdot m + 21 \cdot m + 36\,081 = 780\,000 \, J$$

$$345.8 \cdot m = 743\,919 \, J$$

$$m = 2\,151.3 \, kg$$

- (b) Determine the percentage contribution of the Change in kinetic energy, Rolling resistance work, and Aerodynamic drag work

$$\Delta KE = 324.8 \cdot 2\,151 = 698\,742 \, J \approx 699 \, kJ => 89\%$$

$$W_{\text{roll}} = 21 \cdot 2\,151 = 45\,177 \, J \approx 45 \, kJ => 6\%$$

$$W_{\text{drag}} \approx 36.1 \text{ kJ} \Rightarrow 5\%$$

These calculations demonstrate the physics of vehicle motion, and are meant to provide an intuition for why electric vehicles are significantly more efficient than internal combustion engines. They represent the theoretical minimum energy required for movement; any consumption beyond this reflects mechanical or thermal inefficiencies.

16. e-Ammonia production facility utilizes e-hydrogen generated via water electrolysis. Based on the techno-economic parameters provided below, calculate the Levelized Cost of Fuel (LCOF). Additionally, determine the resulting increase in LCOF if the hydrogen price rises by 20%.

- Capacity – 5 MW_{NH₃}
- Capex – 950 \$/kW
- Fixed opex – 3% of annual capex
- Lifetime – 30 years
- Discount rate – 7%
- Annual operating hours – 7000 hours
- Electricity consumption – 0.13 MWh/MWh_{NH₃}
- Hydrogen consumption – 1.1 MWh_{H₂}/MWh_{NH₃}
- Electricity price – 35 \$/MWh
- Hydrogen price – 95 \$/MWh_{H₂}

$$Output_{\text{NH}_3} = 5\,000 \text{ kW}_{\text{NH}_3} \cdot 7000 \text{ hours} = 35\,000\,000 \text{ kWh}_{\text{NH}_3}$$

$$crf = \frac{7\% \cdot (1 + 7\%)^{30}}{(1 + 7\%)^{30} - 1} = \frac{0.533}{6.61} = 8\%$$

$$CAPEX_{\text{annual}} = 5\,000 \text{ kW}_{\text{NH}_3} \cdot 950\$/\text{kW} \cdot 8\% = 380\,000 \$$$

$$OPEX_{\text{annual}} = 380\,000 \$ \cdot 3\% = 11\,400 \$$$

$$Input_{\text{electricity}} = 35\,000\,000 \text{ kWh}_{\text{NH}_3} \cdot 0.13 \text{ kWh}/\text{kWh}_{\text{NH}_3} = 4\,550\,000 \text{ kWh}$$

$$Input_{\text{H}_2} = 35\,000\,000 \text{ kWh}_{\text{NH}_3} \cdot 1.1 \text{ kWh}_{\text{H}_2}/\text{kWh}_{\text{NH}_3} = 38\,500\,000 \text{ kWh}_{\text{H}_2}$$

$$Cost_{electricity} = 4\,550\, MWh \cdot 35\, \$/MWh = 159\,250\, \$$$

$$Cost_{H_2} = 38\,500\, MWh_{H_2} \cdot 95\, \$/MWh_{H_2} = 3\,657\,500\, \$$$

$$LCOF = \frac{380\,000\, \$ + 11\,400\, \$ + 159\,250\, \$ + 3\,657\,500\, \$}{35\,000\,000\, kWh_{NH_3}} = 120\, \$/MWh_{NH_3}$$

If the hydrogen price is 20% higher

$$Cost_{H_2} = 38\,500\, MWh_{H_2} \cdot 95\, \$/MWh_{H_2} \cdot 120\% = 4\,389\,000\, \$$$

$$LCOF = \frac{380\,000\, \$ + 11\,400\, \$ + 159\,250\, \$ + 4\,389\,000\, \$}{35\,000\,000\, kWh_{NH_3}} = 141\, \$/MWh_{NH_3}$$

These calculations demonstrate how e-fuel costs are driven by hydrogen prices. Since hydrogen cost is driven heavily by electricity prices, these figures illustrate why low-cost electricity is essential for e-fuels.

ENCN423 / 627 Formula Sheet

$$V_{MPP-min} = V_{MPP} \times \left[1 + \left((T_{max} + T_{add} - T_{STC}) \times \frac{T_k}{100} \right) \right]$$

$$V_{OC-max} = V_{OC} \times \left[1 + \left((T_{min} - T_{STC}) \times \frac{T_k}{100} \right) \right]$$

$$E_{\text{useful}} = C \cdot V_{\text{nominal}} \cdot (\text{SoC}_{\text{max}} - \text{SoC}_{\text{min}})$$

$$V_{\text{nominal}} = \frac{V_{\text{max}} + V_{\text{min}}}{2}$$

$$\text{acceleration} = \frac{v_2 - v_1}{t}$$

$$\text{distance} = \frac{v_1 + v_2}{2} \cdot t$$

$$\Delta KE = \frac{1}{2} m (v_2^2 - v_1^2)$$

$$W_{\text{roll}} = C_{\text{roll}} \cdot m \cdot g \cdot d$$

$$W_{\text{drag}} = \frac{1}{8} \cdot \rho \cdot C_{\text{drag}} \cdot A \cdot \frac{v_2^4 - v_1^4}{a}$$

$$LCOF = \frac{CAPEX \cdot crf + OPEX + \sum (LCOI_n \cdot Input_n)}{Output_{\text{fuel}}}$$

$LCOI_n$ – Levelized Cost of Input n (e.g., $LCOE$, $LCOH_2$, cost of CO_2 , heat, etc.).

$$crf = \frac{r \cdot (1 + r)^{\text{Lifetime}}}{(1 + r)^{\text{Lifetime}} - 1}$$

$$C = \frac{E_{\text{total}}}{V_{\text{nominal}}}$$